ISA5810 DM2026 Lab 1 — Master Notebook Questions

Fill in this file as you work through DM2026-Lab1-Master.ipynb. Submit this completed file converted to PDF together with the rest of your Lab 1 work.

Please do not change the style of this document.

Question numbers 1–10 follow the same order as the notebook (top to bottom).

Student Information

Please complete the fields below. For **English Name** and **Chinese Name**, you may fill in **only one** if you do not have the other.

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Instructions

These questions go with the Master Jupyter notebook (DM2026-Lab1-Master.ipynb). Each question is tied to a specific notebook section. Open the notebook, find **Question N (take home)**, and write your answer here.

We **strongly** suggest running the notebook to learn the lab, but the TAs will **not** audit whether you executed every cell to answer the questions in this document.

How to answer each question

1. Find the **notebook section** listed under the question (use the Table of Contents if needed).
2. Read the task; change parameters when asked (e.g. stop_words, thresholds, k, minSup).
3. Write a clear answer in the **Answer** space below each question.
4. Leave the question text unchanged; write only in the answer and disclosure fields.

Sources, honesty, and AI use

Acceptable vs. Unacceptable AI Usage: Generative AI should serve as an interactive tutor to guide your understanding, not as an automated solver completing the assignment for you.

- **Permitted (AI as Guide & Editor):**
  - Asking AI to explain unfamiliar theoretical concepts, discuss algorithmic trade-offs, or troubleshoot runtime errors in the notebook.
  - Polishing the grammar, phrasing, and readability of your own original, self-written drafts.

- **Prohibited (AI as Solver):**
  - Copying assignment questions directly into an AI tool to generate answers.
  - Using AI to fabricate experimental results or observations without executing the code in DM2026-Lab1-Master.ipynb.
  - Submitting machine-generated prose wholesale.

- **External material:** If you use websites, books, papers, or documentation beyond the notebook, cite them with **clickable links** (or DOI/ISBN when no URL exists).

- **When AI was used:** Your answer is acceptable if you (1) **disclose** AI use and (2) include **at least one external link** that validates the information in your answer (official docs, textbook, paper, or reputable tutorial — not the AI chat alone as the only source).

- **When AI was not used:** You may write "No" under **AI used** or leave it blank only if you truly did not use AI for that question. References are still encouraged if you looked anything up.

Under each question, complete:

- **AI used for this answer:** Yes / No
- **If yes, how:** (e.g. brainstorming, grammar, concept explanation)
- **External references (links):** *(required when AI = Yes; encouraged otherwise)*

Write in English. Keep answers concise and specific.

Questions

Question 1: Data Filtering & Domain Relevance

Notebook: Section 3.2 — Question 1 (take home)

Task: Filter the dataframe **X** for the category **sci.med** and check its length.

Questions:

1. How many documents are in this specific category?
2. From a data mining perspective, why is isolating specific categories a crucial first step before applying algorithms like frequent pattern mining?

Answer:

1. Category sci.med has **594** doucuments
2. Without separating the categories, documents from different categories influence each other, so the discovered patterns become mixed and blurred. Isolating each category first lets us identify patterns that are specific to it and clearly distinguish the characteristics of each category.

AI used for this answer: Yes

If yes, how: clarifying the question, explaining unfamiliar terms, grammar polishing

External references (links): https://ithelp.ithome.com.tw/articles/10187244

Question 2: Handling Missing Data

Notebook: Section 4.1 — Question 2 (take home) (after the **dropna()** demonstration)

Task: In Section 4.1, a dummy record with a missing value is injected and then removed using the **dropna()** function.

Question: If 40% of documents were missing the **category_name** label, why would simply dropping these rows (with **dropna()**) be a flawed data mining strategy? Name one objective alternative so the text data can still be utilized.

Answer:

1. Dropping 40% of the documents loses a large amount of information. If the missing rows concentrate in certain categories, the class distribution of the remaining data changes, so the mining results become biased and no longer reflect the original data.
2. Alternative: Estimate Missing Values. If category_name is missing, the numeric category is most likely missing too, so we can train a classifier on the 60% of documents that have labels, and use it to predict the labels of the remaining 40% from their text.

AI used for this answer: Yes

If yes, how: Grammar polishing

External references (links): Section1 and 2 Handout

Question 3: Data Sampling Risks

Notebook: Section 5.1 — Question 3 (take home)

Task: In Section 5.1, the code uses **X_sample = X.sample(n=1000)** to create a subset. Compare the bar chart of the original category distribution to the sampled distribution.

Questions:

1. Do the exact category proportions perfectly match the original?
2. What is the specific risk of simple random sampling on a highly imbalanced dataset?

Answer:

1. It's almost the same.
2. With an imbalanced dataset, simple random sampling may produce a sample that does not reflect the original class distribution. By chance, a minority class can be heavily under-represented or even missed entirely (or, less often, over-represented), so models or patterns learned from the sample can be biased.

AI used for this answer: Yes

If yes, how: clarifying the question, grammar polishing

External references (links): Extra code in written in master.ipynb, https://imbalanced-learn.org/stable/introduction.html, Section1 and 2 Handout

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Question 4: Text Vectorization and Matrix Sparsity

Notebook: Section 5.3 — Question 4 (take home)

Task: In Section 5.3, the **CountVectorizer** converts text into a document-term matrix using **stop_words='english'**. Change this parameter to **stop_words=None** and re-run the cell to check the shape of **X_counts**.

Questions:

1. Does the total number of features (columns) increase or decrease?
2. Since stopwords are highly common words that appear in almost all documents, what effect does adding them back have on the overall sparsity (the percentage of zeros) of the matrix?

Answer:

1. Original: 35482, New:35788. The total number of features increased.
2. It decreases sparsity. Stopwords appear in almost every document, so the added columns are mostly non-zero. Adding these dense columns increases the proportion of non-zero entries and therefore lowers the percentage of zeros in the matrix.

AI used for this answer: Yes

If yes, how: Helping with writing the extra code, grammar polishing

External references (links): The extra code that compute the sparsity of using 'english' and None, to get the true result.

Question 5: Feature Selection — Variance Thresholding

Notebook: Section 5.3.1 — Question 5 (take home)

Task: In the variance filtering section, re-run **variance_filter** using threshold values **threshold = 0.0, threshold = 0.01, and threshold = 0.05**. Record the number of features marked as **keep == True**.

Questions:

1. What happens to the total size of the vocabulary as the variance threshold increases from 0.0 to 0.05?
2. In terms of document frequency, what kinds of words have variance close to zero and are eliminated first?
3. Why is eliminating near-zero variance features considered a safe way to reduce dimensionality without harming model accuracy?

Answer:

1. As the threshold increases from 0.0 to 0.05, the vocabulary shrinks sharply, most of the vocabulary is removed already at a very small threshold.
2. The removed terms with the lowest variance (all 0.000443) appear in only one of the 2,257 documents (document frequency = 1, checked by writing an extra cell). Their counts are zero everywhere else, so they barely vary and are removed first.
3. A feature with near-zero variance behaves almost identically across all samples, so it cannot help separate one sample or class from another. Removing it therefore reduces dimensionality with little effect on model performance.

Threshold	Number of terms kept (keep == True)
0.0	35482
0.01	5089
0.05	1491

AI used for this answer: Yes

If yes, how: Helping with writing the extra code, explaining unfamiliar terms, grammar polishing

External references (links): https://www.youtube.com/watch?v=p2LPBF8DB4c (Master Notebook Guide provided by TAs), the extra cell's output

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Question 6: Feature Correlation — Linear (Pearson) vs. Monotonic (Spearman)

Notebook: Section 5.3.2 — Question 6 (take home)

Task: Run the correlation filter against the binary indicator for the **comp.graphics** class. Inspect the top 10 features sorted by **pearson_r** and compare them to the top 10 features sorted by **spearman_r**.

Questions:

1. Are the top-correlated words generic stop words or domain-specific terminology for graphics?
2. What is the mathematical difference in what **Pearson** correlation measures versus what **Spearman** rank correlation measures?
3. In what specific scenario would a word have a high **Spearman** rank correlation with a topic, but a significantly lower **Pearson** correlation?

Answer:

1. The Pearson top 10 contains a mix of generic words (e.g., thanks, hi, writes, looking) and a few graphics terms (e.g., vga, video, card). In contrast, most of the Spearman top 10 are graphics-related terms .
2. Pearson correlation measures the strength of a linear relationship between two variables using their raw values, so a non-linear relationship weakens it. Spearman correlation first converts the values to ranks and then computes the Pearson correlation on those ranks, so it measures whether one variable consistently increases (or decreases) as the other does, even if it is not a straight line.
3. A word would have a high Spearman but a lower Pearson correlation when its relationship with the topic is monotonic but not linear. For example, a term's count tends to be higher in documents of the topic, but a few documents contain extremely large counts. These extreme values distort the straight-line fit, lowering Pearson, while the rank order stays consistent, so Spearman remains high.

AI used for this answer: Yes

If yes, how: Clarifying the question, grammar polishing, discussing information in the output result, translating some hard words

External references (links):
https://docs.scipy.org/doc/scipy/reference/generated/scipy.stats.pearsonr.html

https://docs.scipy.org/doc/scipy/reference/generated/scipy.stats.spearmanr.html

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Question 7: Frequent Pattern Mining — Algorithm & Parameter Dynamics

Notebook: Section 5.4.2 — Question 7 (take home)

Task: In the Frequent Pattern Mining section, run **Top-K FAE** with **k = 100 and k = 500**. Then, run **MaxFPGrowth** using minimum support thresholds **minSup=9 and minSup= 3**.

Questions:

1. How does decreasing **minSup** (from 9 to 3) in **MaxFPGrowth** affect the execution runtime and the total number of discovered patterns?
2. How does the parameter k in **Top-K FAE** differ fundamentally from **minSup** in **MaxFPGrowth** when deciding which patterns to keep?
3. Why are "maximal" frequent patterns structurally more concise than standard frequent patterns?

Answer (1) — Runtime and pattern count when lowering minSup:

	Execution runtime	Total number
minSup=3	0.9102075099945068	9547
minSup=9	0.04603886604309082	710
	About 19.8 times	About 13.4 times

Decreasing minSup from 9 to 3 increased the runtime and the number of patterns. This indicates that many patterns occur only 3–8 times, and since more patterns qualify as frequent, the algorithm takes longer to mine.

Answer (2) — Top-K k vs MaxFPGrowth minSup:

k in Top-K FAE fixes the number of patterns: it keeps the k patterns with the highest support, regardless of how often they occur, so the minimum support varies with the data. In contrast, minSup in MaxFPGrowth fixes a frequency threshold: it keeps every pattern that occurs at least minSup times, so the number of patterns varies with the data.

Answer (3) — Maximal vs standard frequent patterns:

Because every subset of a frequent pattern is also frequent, standard mining lists all of them, but they repeat the same information. Maximal mining removes these subsets to avoid redundancy, so the result is more concise.

AI used for this answer: Yes

If yes, how: Clarifying the question, explaining unfamiliar terms, grammar polishing

External references (links): https://www.youtube.com/watch?v=p2LPBF8DB4c (Master Notebook Guide provided by TAs), PAMI introduction in Master.ipynb

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Question 8: Dimensionality Reduction — Motivation

Notebook: Section 5.5 — Question 8 (take home) (before PCA/t-SNE code)

Task: After text vectorization, the **X_counts** matrix creates tens of thousands of distinct features (words).

Questions:

1. If we want to visually map how these documents group by category on a scatter plot, why is it mathematically and visually impossible to plot the raw **X_counts** matrix directly?
2. What specific category of data mining algorithms (referenced in the helper files) must be applied first?

Answer:

1. A scatter plot can show at most 2–3 axes, but X_counts has tens of thousands of features. This cannot be drawn in 2D/3D, and humans cannot perceive such a high-dimensional space, so the raw matrix cannot be plotted directly.
2. Dimensionality reduction. It projects the tens of thousands of features into 2 or 3 dimensions so that the documents can be plotted.

AI used for this answer: Yes

If yes, how: Clarifying the question, grammar polishing

External references (links): https://www.youtube.com/watch?v=p2LPBF8DB4c (Master Notebook Guide provided by TAs), Section 1 and 3 handout, Master.ipynb

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Question 9: Dimensionality Reduction — Linear (PCA) vs. Non-linear Manifolds (t-SNE/UMAP)

Notebook: Section 5.5 — Question 9 (take home) (after 2D/3D plots)

Task: In the 2D/3D Dimensionality Reduction section, compare the scatter plot projection generated by **PCA** against that generated by **t-SNE** or **UMAP**.

Questions:

1. Which method **(PCA or t-SNE/UMAP**) shows distinct, separated cluster islands for the 4 document categories, and which shows heavy overlap around the center?
2. Why does **PCA** struggle to separate high-dimensional text categories compared to **t-SNE** or **UMAP**?
3. Why must data analysts be careful not to over-interpret the physical distance between separate clusters in a **t-SNE** plot?

Answer:

1. PCA shows heavy overlap: almost all documents collapse into a small region near the origin, and the soc.religion.christian category is barely visible. UMAP separates the categories best: documents of the same category lie close together, forming distinguishable regions, although the boundaries between them remain blurry in the center.
2. PCA is a linear projection, so if the category structure in the high-dimensional space is non-linear, the projection cannot preserve it, and the categories end up overlapping. In contrast, t-SNE and UMAP are non-linear and can capture such structure.
3. t-SNE only preserves the relative distances between nearby points from the high-dimensional space, so distances within a cluster are meaningful. Distances between different clusters (non-neighbors) are not guaranteed to match the original data, so they should not be interpreted as real differences.

AI used for this answer: Yes

If yes, how: Clarifying the question, discussing the plot, grammar polishing

External references (links): https://www.youtube.com/watch?v=p2LPBF8DB4c (Master Notebook Guide provided by TAs), Section 1 and 3 handout, Knowledge learned from Introduction to machine learning

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Question 10: One-Hot Binarization

Notebook: Section 5.6 — Question 10 (take home)

Task: In Section 3.1, examine **X['category']** (integers 0, 1, 2, 3). Then look at the output of **LabelBinarizer** in Section 5.6, where each category is converted into a binary vector (e.g., [1, 0, 0, 0]).

Questions:

1. How many binary columns are created to represent the 4 categories?
2. If an algorithm (like linear regression or KNN) is given the raw integer column **(0, 1, 2, 3)** as input, what false mathematical assumption does it make about the categories?
3. How does one-hot binarization eliminate this false assumption?

Answer:

1. Four. For example, category 1 corresponding to [0, 1, 0, 0]
2. Given raw integers (0, 1, 2, 3), algorithms like KNN or linear regression wrongly assume that the categories have a numeric order and magnitude. For example, class 3 is treated farther from class 0 than class 1 is, although the integers are just arbitrary labels.
3. One-hot binarization represents each category as its own binary vector, so every pair of categories has the same distance ($\sqrt{2}$). This removes any false ordering or distance between the categories.

AI used for this answer: Yes

If yes, how: Grammar polishing

External references (links): Knowledge learned from Introduction to machine learning

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